SALBUTAMOL

Reliever inhaler

Hypokalaemia at high doses

CHEMICAL NAME (RS)-4-[2-(tert-butylamino)-1-hydroxyethyl]-2-(hydroxymethyl)phenol (sulfate); also called albuterol

CLASS Short-acting β2-adrenoceptor agonist (SABA) bronchodilator

TRADE NAMES Ventolin · Proventil · Asthalin · Airomir

THERAPEUTIC USES / INDICATIONS

- Relief of acute bronchospasm in asthma and COPD.
- Acute severe asthma (nebulised or IV).
- Prevention of exercise-induced bronchospasm.
- Hyperkalaemia (nebulised, adjunct).

PHARMACODYNAMICS

- Stimulates **β2 receptors** on bronchial smooth muscle → ↑ cAMP → bronchodilation.
- Drives K^+ into cells (hypokalaemia); relaxes uterus.

ADMINISTRATION

- **Inhaler (pMDI):** shake, exhale, seal lips, press and inhale slowly, hold breath 10 s; use a spacer (children, acute attacks).
- **Nebuliser:** with oxygen in acute asthma (air in COPD with CO_2 retention).
- **IV** infusion in life-threatening asthma (ICU/HDU).
- **PO** syrup/tablets rarely used (more side effects).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Fine tremor, palpitations, tachycardia.
- Headache, nervousness.
- Muscle cramps.

ADVERSE EFFECTS (SERIOUS)

- Hypokalaemia (high doses, with steroids, theophylline, diuretics).
- Arrhythmias, myocardial ischaemia.
- Lactic acidosis (IV/nebulised high dose).
- Paradoxical bronchospasm.

DOSAGES

- **Inhaler:** 100–200 mcg (1–2 puffs) as needed; up to 4 times daily.
- **Acute asthma (spacer):** up to 10 puffs, repeat every 20 min.
- **Nebulised:** 2.5–5 mg every 20–30 min or back-to-back in severe attack (child <5 yr 2.5 mg).
- **IV (severe bronchospasm):** usually 3–20 mcg/min (10 mcg/mL solution); higher in respiratory failure.
- **Hyperkalaemia:** 10–20 mg nebulised.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** cardiovascular disease, arrhythmias, hyperthyroidism, diabetes (IV → hyperglycaemia/ketoacidosis), hypokalaemia.

KEY INTERACTIONS

- **Non-selective β-blockers (propranolol):** block effect, bronchospasm.
- **Corticosteroids, diuretics, theophylline:** ↑ hypokalaemia.
- **MAOIs, TCAs:** ↑ cardiovascular effects.

PHARMACOKINETICS

- **Onset:** inhaled 5 min; peak 30 min. **Duration:** 4–6 h.
- Metabolised in liver (sulfate conjugation).
- Excreted in urine. **t½:** 4–6 h.

NURSING MANAGEMENT

- Assess RR, SpO_2, wheeze, peak flow before and after.
- Check inhaler/spacer technique.
- Monitor HR, K^+ (high doses), glucose (IV).
- Increasing reliever use = poor control → review.
- Nebuliser: correct driving gas, mask fit.

PATIENT EDUCATION

- Carry your reliever everywhere.
- If you need it more than twice a week, see your doctor (asthma not controlled).
- Tremor and fast heartbeat are common after use.
- Rinse and dry the spacer weekly; check dose counter.

KEY POINTS

- SABA reliever — onset 5 min, lasts 4–6 h.
- Tremor, tachycardia, hypokalaemia.
- Spacer improves delivery.
- Overuse = poorly controlled asthma.

SOURCES CHECKED **[446]** Ventolin Solution for IV Infusion (UK SmPC) **[447]** BTS/SIGN asthma guideline (SIGN 158) **[448]** UKKA hyperkalaemia guideline (2023)

Full references p. 432–457